School Worksheet

Class 12 Mathematics, Chapter 10: Vector Algebra

Maximum marks: **47** **20** questions

Name _____ Date _____ Score _____ / 47

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

The unit vector perpendicular to the vectors $\hat{i} - \hat{j}$ and $\hat{i} + \hat{j}$ is:

(A) $\hat{k}$

(B) $-\hat{k} + \hat{j}$

(C) $\dfrac{\hat{i} - \hat{j}}{\sqrt{2}}$

(D) $\dfrac{\hat{i} + \hat{j}}{\sqrt{2}}$

2 MCQ — 1 mark

If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = \sqrt{37}$, $|\vec{b}| = 3$ and $|\vec{c}| = 4$, then the angle between $\vec{b}$ and $\vec{c}$ is:

(A) $\dfrac{\pi}{6}$

(B) $\dfrac{\pi}{4}$

(C) $\dfrac{\pi}{3}$

(D) $\dfrac{\pi}{2}$

3 MCQ — 1 mark

The value of λ for which $2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular is:

(A) $\dfrac{5}{2}$

(B) $-\dfrac{5}{2}$

(C) 2

(D) $\dfrac{3}{2}$

4 MCQ — 1 mark

If $|\vec{a}| = 2$, $|\vec{b}| = 5$ and $\vec{a} \cdot \vec{b} = 8$, then $|\vec{a} \times \vec{b}|$ is:

(A) 2 **(B)** 4

(C) 6 **(D)** 8

Q No.	Question	Marks
5	MCQ The projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ is: **(A)** $\frac{10}{\sqrt{17}}$ **(B)** $\frac{5\sqrt{6}}{3}$ **(C)** $\frac{10}{6}$ **(D)** $\sqrt{6}$	1 mark
6	Assertion-Reason **Assertion (A):** The vectors $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ are collinear. **Reason (R):** Two vectors $\vec{a}$ and $\vec{b}$ are collinear if $\vec{b} = \lambda\vec{a}$ for some scalar λ. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Find the area of the parallelogram whose adjacent sides are $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.	2 marks

Level 2

16 marks

8	Short answer Find a vector of magnitude 21 units in the direction opposite to that of $\overrightarrow{AB}$, where A and B are the points $A(2, 1, 3)$ and $B(8, -1, 0)$ respectively.	2 marks
9	Short answer Find a unit vector perpendicular to both $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.	2 marks
10	Short answer Two friends flying kites from different locations find the strings of their kites crossing each other. The strings can be represented by the vectors $\vec{a} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + 4\hat{k}$. Determine the angle formed between the kite strings. (Assume there is no slack in the strings.)	2 marks
11	Short answer Find the area of the triangle with vertices $A(1, 1, 2)$, $B(2, 3, 5)$ and $C(1, 5, 5)$.	3 marks
12	Short answer Find the projection of the vector $(\vec{b} + \vec{c})$ on the vector $\vec{a}$, where $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{k}$.	3 marks

Q No.	Question	Marks
13	Short answer If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j}$, find λ such that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$.	2 marks
14	Short answer Find the position vector of the point R which divides the line joining $P(2\vec{a} + \vec{b})$ and $Q(\vec{a} - 3\vec{b})$ externally in the ratio $1 : 2$.	2 marks
Level 3		**23 marks**
15	Short answer If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$, prove that $\vec{a} = \pm 2(\vec{b} \times \vec{c})$.	3 marks
16	Long answer Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c} \cdot \vec{d} = 15$.	5 marks
17	Short answer Find a vector of magnitude 5 units perpendicular to each of $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$, where $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.	3 marks
18	Short answer Using vectors, find the area of the triangle ABC with vertices $A(1, 2, 3)$, $B(2, -1, 4)$ and $C(4, 5, -1)$.	3 marks
19	Long answer Show that the points with position vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$ are the vertices of a right-angled triangle. Also find the remaining angles' cosines.	5 marks
20	Case study **Case Study:** A toy drone starts from the origin O. It first moves to point A with $\overrightarrow{OA} = 2\hat{i} + 3\hat{j} + \hat{k}$ and is then commanded to point B with $\overrightarrow{OB} = 4\hat{i} + \hat{j} + 3\hat{k}$ (units in metres). **(i)** Find the displacement vector $\overrightarrow{AB}$. 1 mark **(ii)** Find the distance travelled from A to B and the unit vector along $\overrightarrow{AB}$. 1 mark **(iii)** Find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. 2 marks	4 marks

Answer key

Chapter 10: Vector Algebra, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $\hat{k}$
2	(C) $\frac{\pi}{3}$
3	(A) $\frac{5}{2}$
4	(C) 6
5	(B) $\frac{5\sqrt{6}}{3}$
6	(A) Both A and R are true and R is the correct explanation of A
7	Area $= 2\sqrt{2}$ sq units

Level 2

8	$-18\hat{i} + 6\hat{j} + 9\hat{k}$
9	$\pm\frac{1}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$
10	$\theta = \cos^{-1}\left(\frac{\sqrt{21}}{7}\right)$
11	Area $= \frac{\sqrt{61}}{2}$ sq units
12	Projection $= 4$
13	$\lambda = 8$
14	$\overrightarrow{OR} = 3\vec{a} + 5\vec{b}$

Level 3

15	$\vec{a} = \pm 2(\vec{b} \times \vec{c})$, proved
16	$\vec{d} = \frac{1}{3}(160\hat{i} - 5\hat{j} - 70\hat{k})$
17	$\pm\frac{5}{\sqrt{6}}(-\hat{i} + 2\hat{j} - \hat{k})$, i.e. $\pm\frac{5\sqrt{6}}{6}(-\hat{i} + 2\hat{j} - \hat{k})$
18	Area $= \frac{\sqrt{274}}{2}$ sq units
19	Right angle at C; $\cos A = \sqrt{\frac{35}{41}}$, $\cos B = \sqrt{\frac{6}{41}}$
20	(i) $2\hat{i} - 2\hat{j} + 2\hat{k}$; (ii) $2\sqrt{3}$ m, $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$; (iii) $\cos^{-1}\sqrt{\frac{7}{13}}$